

Experiment 5: Fuel Efficiency Analysis Using NumPy

AIM

To calculate the average fuel efficiency of car models and determine the percentage improvement in fuel efficiency between two cars using NumPy arrays.

ALGORITHM

1. Import the NumPy library.
2. Create a NumPy array containing the fuel efficiency of different car models.
3. Calculate the average fuel efficiency using `np.mean()`.
4. Select the fuel efficiency values of two cars.
5. Calculate the percentage improvement using the formula.
6. Display the average fuel efficiency and percentage improvement.

CODE

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40])

average = np.mean(fuel_efficiency)

car1 = fuel_efficiency[0]

car2 = fuel_efficiency[1]

improvement = ((car2 - car1) / car1) * 100

print("Average fuel efficiency:", average, "mpg")

print("Percentage improvement:", improvement, "%")
```

INPUT

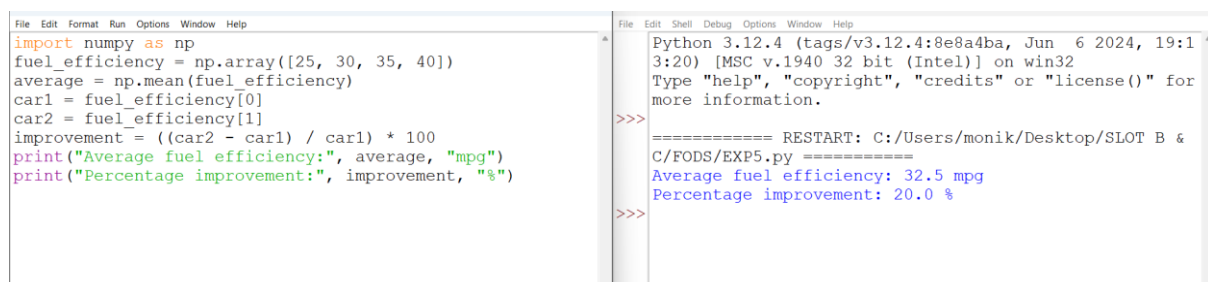
Car 1 = 25 mpg

Car 2 = 30 mpg

Car 3 = 35 mpg

Car 4 = 40 mpg

OUTPUT



The screenshot displays a Python IDE with two windows. The left window shows the code from the 'CODE' section, and the right window shows the output. The output includes the Python version, date, and time, followed by the results of the calculations: 'Average fuel efficiency: 32.5 mpg' and 'Percentage improvement: 20.0 %'.

```
File Edit Format Run Options Window Help
import numpy as np
fuel_efficiency = np.array([25, 30, 35, 40])
average = np.mean(fuel_efficiency)
car1 = fuel_efficiency[0]
car2 = fuel_efficiency[1]
improvement = ((car2 - car1) / car1) * 100
print("Average fuel efficiency:", average, "mpg")
print("Percentage improvement:", improvement, "%")

Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:13:20) [MSC v.1940 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
===== RESTART: C:/Users/monik/Desktop/SLOT B & C/FODS/EXP5.py =====
Average fuel efficiency: 32.5 mpg
Percentage improvement: 20.0 %
>>>
```

RESULT

Thus, the average fuel efficiency was calculated as 32.5 mpg, and the percentage improvement from Car 1 to Car 2 was 20%.